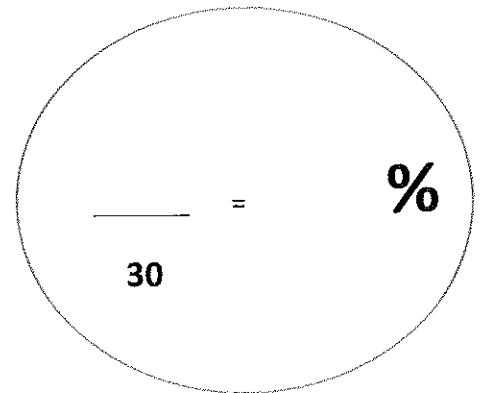


Investigation 1 – Validation Test
Angles and parallel lines

Name: SOLUTIONS

Time allowed: 30 minutes

Calculators Allowed
NO Notes



Answer in the spaces provided.

Show all working clearly, in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. Correct answers given without supporting reasoning may not be awarded full marks.

Question 1

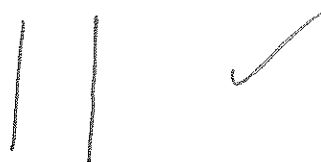
(2 marks)

Draw two sets of parallel lines below, labelled set A and set B.

SET A



SET B



Question 2

(1, 1, 2 = 4 marks)

Consider the line with equation $y=2x+5$.

a) State the gradient of the line above.

2 ✓

b) State the y-intercept of the line above.

5 ✓

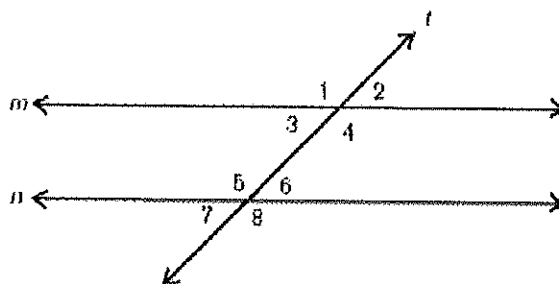
c) What is the equation of the line parallel to this line which has y-intercept of (0,8)?

$y = 2x + 8$
✓ ✓

Question 3

(1, 1, 1, 1, 1, 2, 2 = 9 marks)

Consider the lines below.



a) What do we know about the gradients (slopes) of lines m and n?

same ✓

b) What is the technical name of line t, which cuts lines m and n?

transversal ✓

c) What is the term for the relationship between angles 3 and 6?

alternate ✓

d) What is the term for the relationship between angles 4 and 6?

co-interior ✓

e) What is the term for the relationship between angles 2 and 6?

corresponding ✓

e) If angle 5 is 120° , how many degrees are in angle 3?

$$180^\circ - 120^\circ = 60^\circ$$

✓ ✓

e) If angle 7 is 40° , how many degrees are in angle 4?

supplementary: $\angle 8 = 180^\circ - 27 = 180^\circ - 40^\circ = 140^\circ$ ✓

corresponding: $\angle 8 = \angle 4 = 140^\circ$ ✓

Question 4

(1, 1, 2, 2, 2 = 8 marks)

Figure A

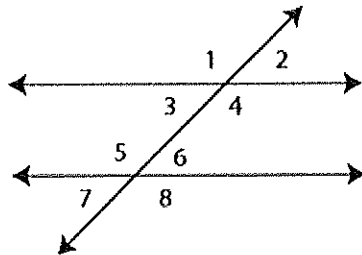
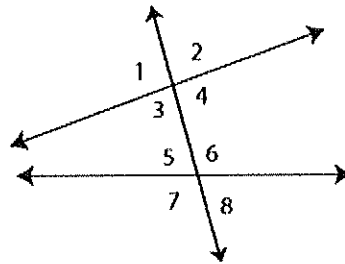


Figure B



In BOTH figure A and figure B, angle 1 is supplementary to angle 2.

Angle 4 is also supplementary to angle 2.

Supplementary angles always add up to a certain number of degrees, let us call this x°

- a) How many degrees are in x° ? (Consider how many degrees there are in a straight line.)

$$180^\circ \checkmark$$

- b) In figure B, if angle 6 is 140° , how many degrees are in angle 8?

$$180^\circ - 140^\circ = 40^\circ \checkmark$$

In figure A, angles 1 and 4 are **vertically opposite**. So are angles 6 and 7.

- a) If angle 1 is 110° , how many degrees are in angle 2?
What do you then know about how many degrees are in 4?

$$\angle 2 = 70^\circ \checkmark$$

$$\angle 4 = 110^\circ \checkmark$$

- b) If angle 6 is 30° , how many degrees is angle 8? How about angle 7?

$$\angle 8 = 150^\circ \checkmark$$

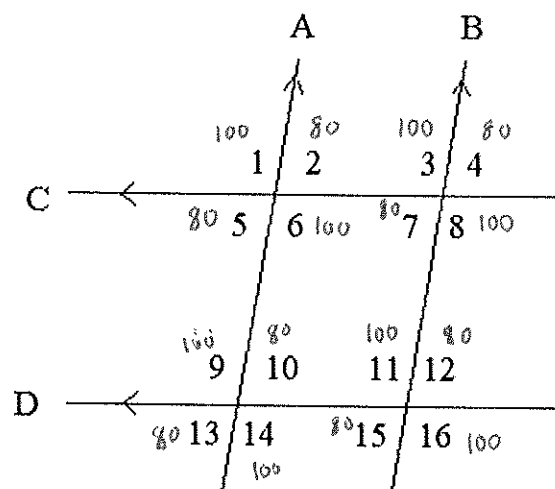
$$\angle 7 = 30^\circ \checkmark$$

- c) Comparing pairs of angles 1 and 4 and angles 6 and 7 in part a) and b), if we know one angle, what do you think we know about the other vertically opposite one?

same $\checkmark \checkmark$

Question 5

(1/2 mark each = max 7 marks)



A company wants to design a mosaic using parallelograms. They decide to use the pattern above, but they need to know all the angles involved for the designer. The only piece of information they have is that angle 1 is 100° . Using your knowledge of parallel lines cut by transversals, determine angles 2-16 and write your answers below:

Number	Angle	Number	Angle	Number	Angle
2	80 ✓	7	80 ✓	12	80 ✓
3	100	8	100	13	100 ✓
4	80 ✓	9	100 ✓	14	100 ✓
5	80	10	80	15	80 ✓
6	100	11	100 ✓	16	100

END OF INVESTIGATION